

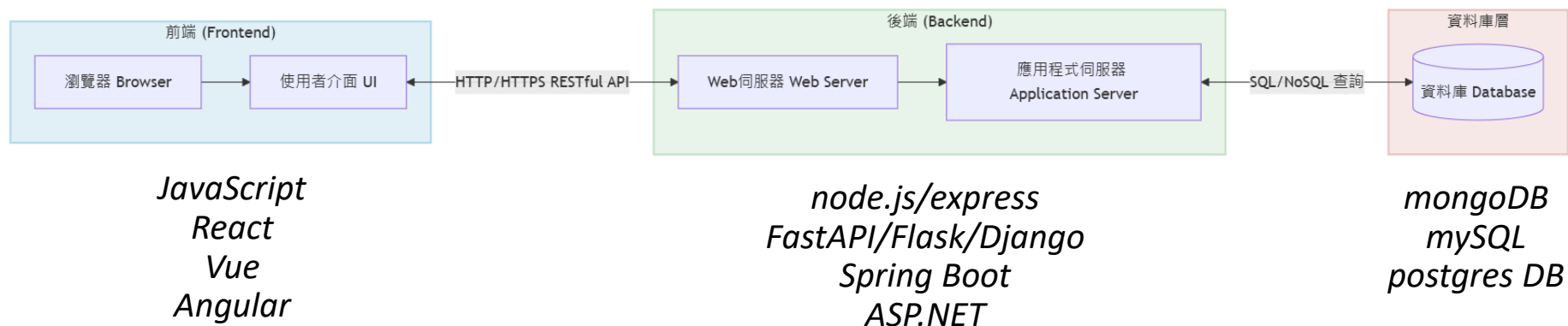


Ajax and Fetch

NTOU CSE
2025

Frontend and Backend

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在程式碼層級可區分為前端與後端，在部署上前後端可以完全分離 (須設定好跨網域存取)，或是屬於同一個專案底下。

Why Backend?₁

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- Data Management
 - ▣ Connect and interact with databases
 - ▣ Ensure data integrity and security
- System Integration
 - ▣ Bridge modern web apps with legacy systems
 - ▣ Facilitate API connections with third-party services
- User Interaction
 - ▣ Enable real-time communication between users
 - ▣ Manage user authentication and authorization

Why Backend?₂

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□ Business Logic

- ▣ Implement and protect core application functionality
- ▣ Centralize critical operations for consistency and maintainability

□ Asynchronous Processing

- ▣ Handle background tasks and scheduled jobs
- ▣ Push notifications and updates to users

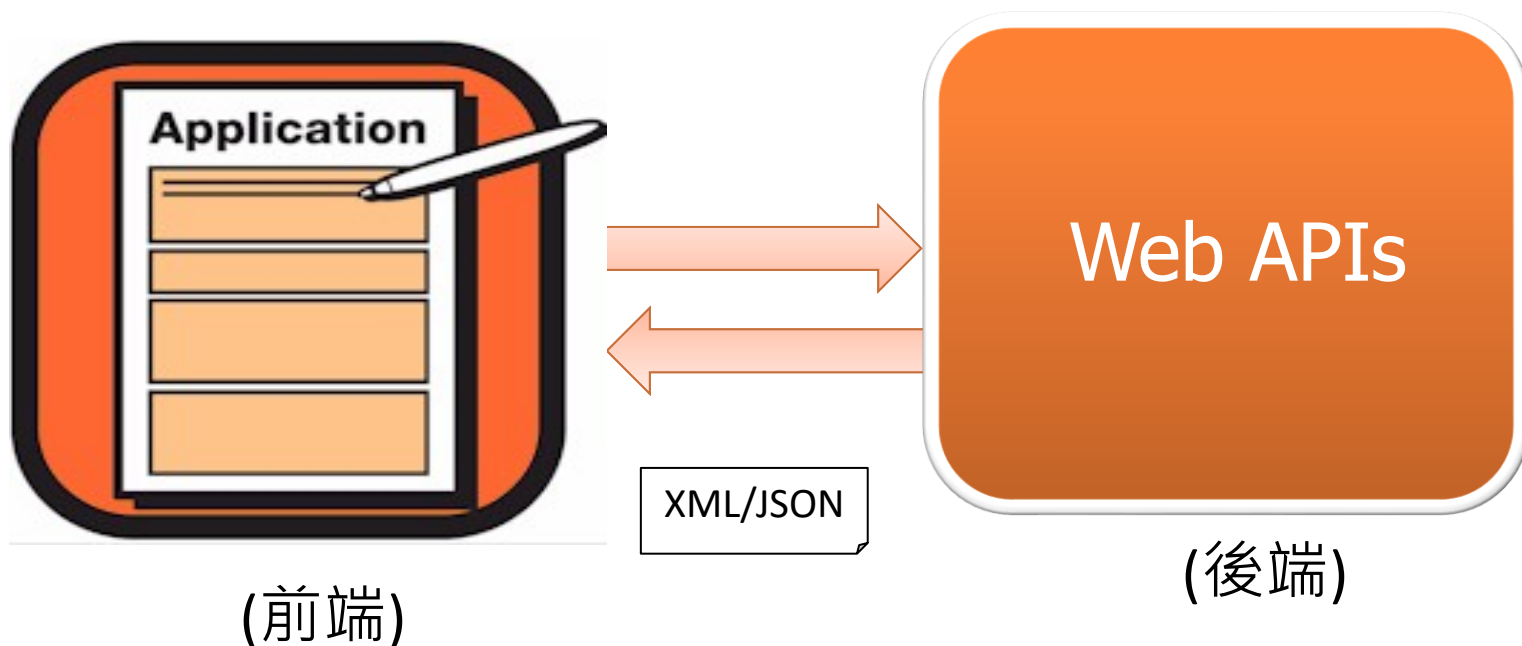
□ Scalability and Performance

- ▣ Optimize server-side operations for improved response times
- ▣ Manage server resources to handle increasing loads

Web API

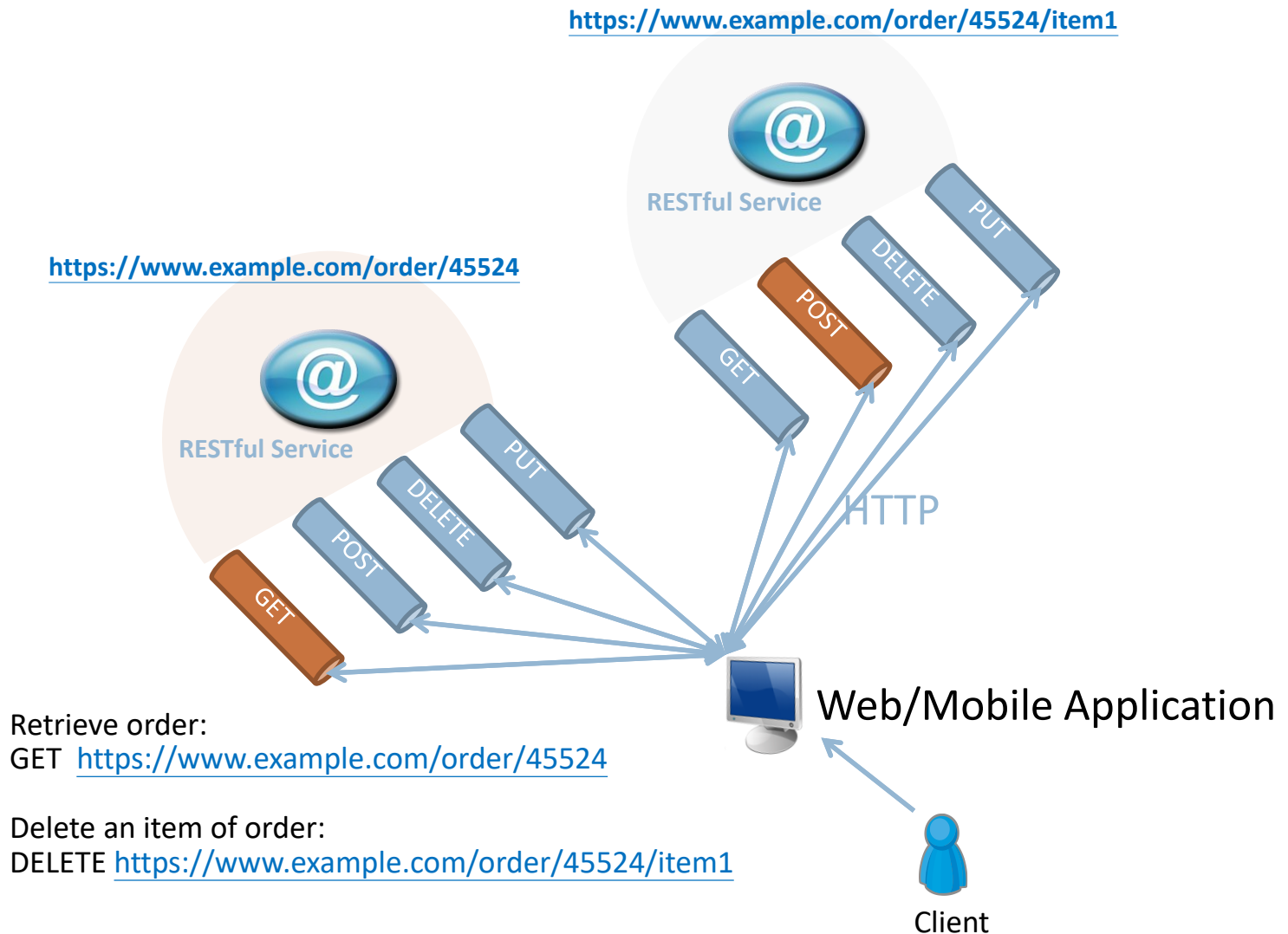
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- A Web API is a development in Web services where emphasis has been moving to simpler **REST**-based communications. (also called **RESTful service**)



RESTful Web Service

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HTTP Methods

(Cornerstone of RESTful Services)

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- **GET**: to get the thing (resource / file) at the requested URL. (讀取)
- **POST**: to ask the server to accept the body info attached to the request, and give it to the thing at the requested URL. (新增)
- **PUT**: to put the enclosed info (the body) at the requested URL. (更換)
- **Patch**: to apply partial modifications to a resource. (更新)
- **DELETE**: to delete the thing (resource / file) at the requested URL. (刪除)

HTTP GET Example

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The “?” separates the path and the parameters.

The amount of data you can send along with the GET is limited.

It’s exposed up here in the browser bar for everyone to see.

Please try:
Chrome: [F12] → Network

HTTP POST Example

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<https://mail.ntou.edu.tw>



帳號account :

albert

@mail.ntou.edu.tw



密碼password :

.....

語言language :

Personal Predefined Config.



記住帳號



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[\[使用說明\]](#)

<https://mail.ntou.edu.tw/>

The Difference between GET and POST

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POST /select/selectBeerTaste.jsp

HTTP/1.1

Host: www.wickedlysmart.com

User-Agent: Mozilla/5.0 (Macintosh; U; PPC Mac OS X Mach-O; en-US; rv:1.4) Gecko/20030624 Netscape/7.1

Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,*/*;q=0.1

Accept-Language: en-us,en;q=0.5

Accept-Encoding: gzip,deflate

Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7

Keep-Alive: 300

Connection: keep-alive

color=dark&taste=malty

Difference between GET and POST

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- With GET, **the parameter data is limited** to what you can stuff into the Request line. (GET參數資料量有限制)
- When you use GET, the parameter data shows up in the browser's input bar. (GET參數內容會出現在網址列)
- GET requests **can be bookmarked**; POST requests cannot. (GET可加入書籤)
- GET is meant to be used for getting things, whereas POST is meant to be used for sending data to be processed. (use the data from the POST body to change something on the server.)

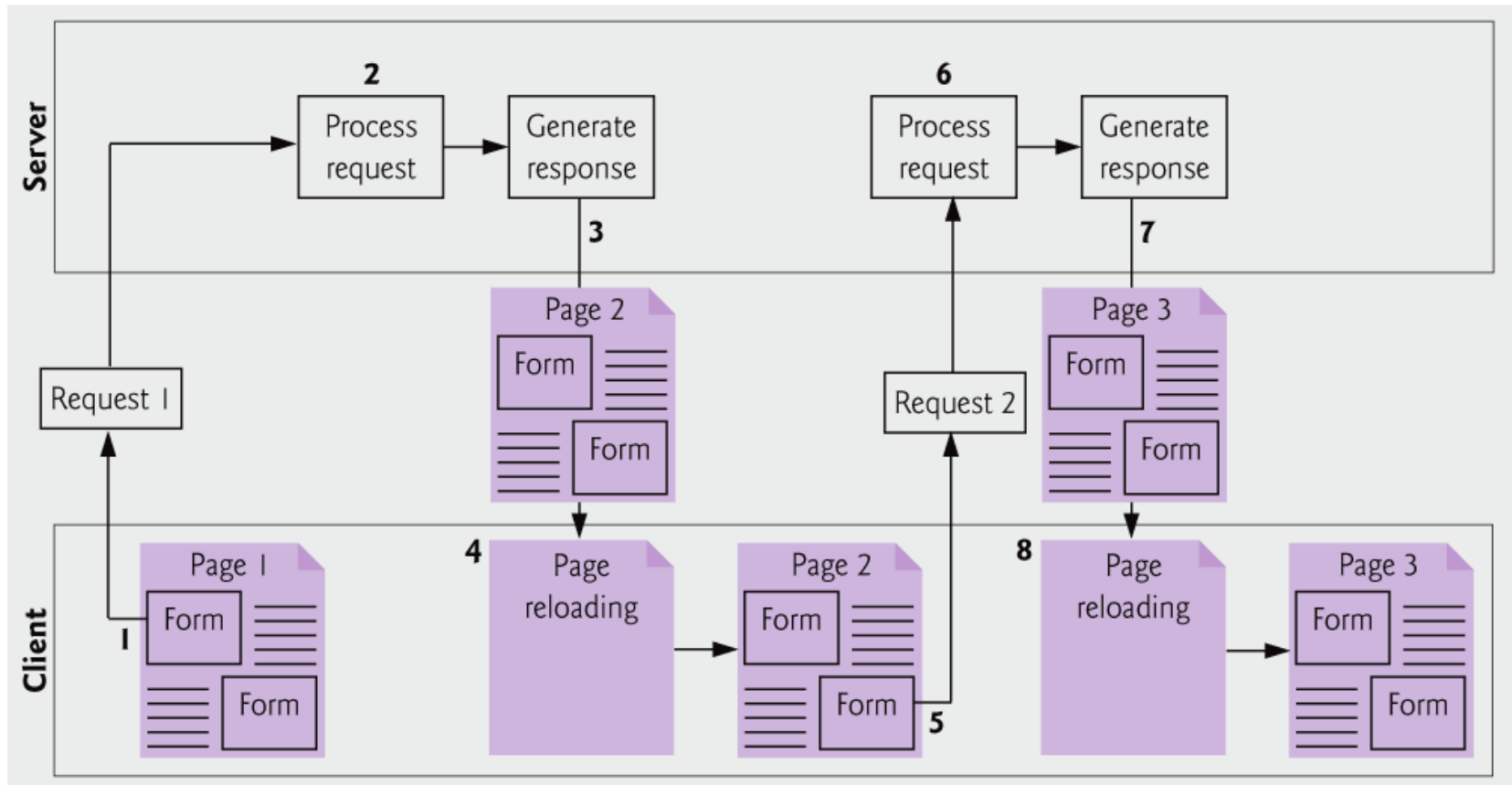
What is AJAX

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- Ajax (pronounced /'eɪdʒæks/)
 - ▣ Shorthand for Asynchronous JavaScript and XML
 - ▣ A group of interrelated web development techniques used on the client-side to create interactive web applications.
- Ajax uses JavaScript to interact directly with the web server, avoiding the submit/response cycle all too familiar to web users.
 - ▣ Ajax **adds server interaction, but without the Submit button.**
 - ▣ Whenever data is needed, the JavaScript in the web page makes a request, and the server replies with data **but not another HTML page.**

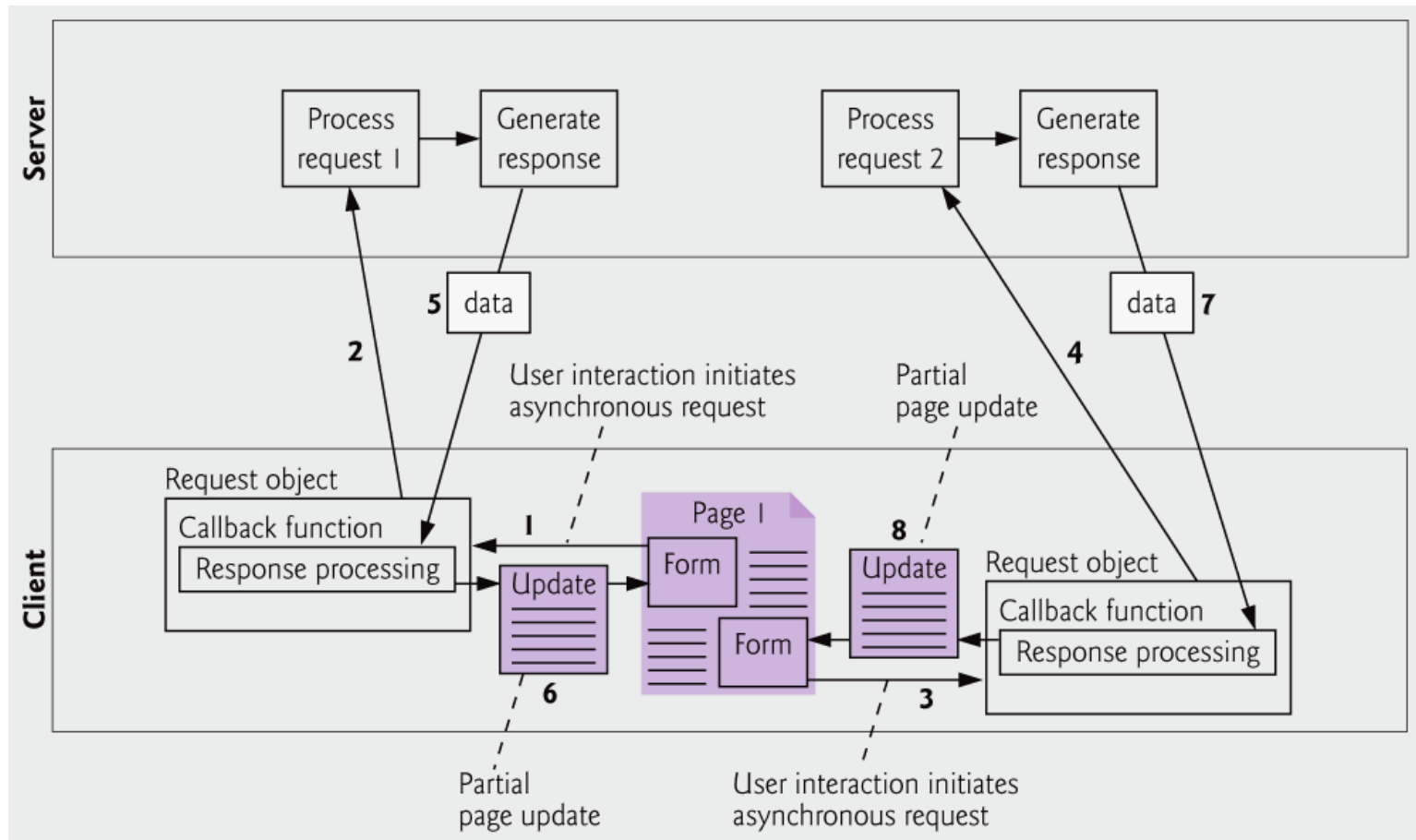
Classic web application reloading the page for every user interaction

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Ajax-enabled web application interacting with the server asynchronously

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Ajax Using Pure JavaScript₁

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- https://www.w3schools.com/js/tryit.asp?filename=tryjs_ajax_database

```
const xhttp = new XMLHttpRequest();
xhttp.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
        document.getElementById("txtHint").innerHTML =
            this.responseText;
    }
};
xhttp.open("GET", "getcustomer.php?q="+str, true);
xhttp.send();
```

Ajax Using Pure JavaScript₂

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```
req.open("GET", url, true);
```

- **"Get"**: The first parameter tells JavaScript whether to submit the request to the server **using HTTP GET or HTTP POST**.
- **url**: The second parameter is the URL we're passing to the server. We created that URL earlier in the method.
- **true**: The last parameter determines **whether or not the call is asynchronous (非同步呼叫)**.
 - ▣ When this parameter is true, the request is sent asynchronously.
 - ▣ When designing Ajax applications, **you always want to set the asynchronous flag to TRue**; basically, it means "don't stop anything, just notify me when the data comes back."

Ajax Using Pure JavaScript₃

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```
req.onreadystatechange = function() {...};
```

- It tells the *req* object to call the **callback** function **whenever a state transition occurs**.
- What Is a Callback (回呼函数)?
 - ▣ A callback is any executable code that is passed as an argument to other code, which is expected to call back (execute) the argument at a given time.
 - ▣ The JavaScript code generates a request that is sent to a servlet.
 - ▣ **When the server (node.js, FastAPI, etc.) returns with the information**, the callback function is invoked; in turn, the callback function can **display the new information to the user**.

Ajax Using Pure JavaScript₄

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`req.send();`

- It sends the request.
- When sending data with POST/PUT, you can include a query string, a data stream, XML, JSON, and so on.
 - ▣ In basic cases, a query string is used to send the data.
 - ▣ For more complex requests, XML or **JSON** formats can be used.

Fetch₁

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- The Fetch API provides a JavaScript interface for accessing and manipulating parts of the **HTTP pipeline**, such as requests and responses. (From ES6)
- It provides a global **fetch()** method that provides an easy, logical way to fetch resources asynchronously across the network.
- This kind of functionality was previously achieved using XMLHttpRequest.

https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch

Fetch₂

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// 最簡單形式

```
fetch('https://api.sampleapis.com/coffee/hot');
```

// 處理回應

```
fetch('https://api.sampleapis.com/coffee/hot')  
  .then(response => response.json()) // 解析 JSON  
  .then(data => console.log(data))   // 使用資料  
  .catch(error => console.log('Error:', error)); // 錯誤處理
```

// async/await 語法

```
async function getData() {  
  try {  
    const response = await fetch('https://api.sampleapis.com/coffee/hot');  
    const data = await response.json();  
    return data;  
  } catch (error) {  
    console.error('Error:', error);  
  }  
}
```

<https://codepen.io/Shang-Pin-Ma/pen/emOOrwR>

Promise

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- A Promise is used to handle asynchronous operations, providing a cleaner and more elegant way to deal with results and errors than traditional callback functions.
- Each Promise has three possible states:
 - ▣ pending – the operation is still in progress
 - ▣ fulfilled – the operation completed successfully
 - ▣ rejected – the operation failed

async/await

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- async/await is syntax sugar built on top of Promises that makes asynchronous code easier to read and write.
- An async function always returns a Promise.
 - ▣ The await keyword pauses the execution of the async function **until the Promise is settled (fulfilled or rejected)**.
 - ▣ Errors in asynchronous code should be handled with **try/catch** when using await.

async/await: Example

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```
const handleGet = async () => {  
    setStatus("載入中...");  
    try {  
        const response = await fetch(BASE_URL + "/posts?_limit=5");  
  
        if (!response.ok) {  
            throw new Error("HTTP 狀態碼：" + response.status);  
        }  
  
        const data = await response.json(); // 等待 JSON 轉換  
        setStatus("GET 成功 (詳情請看 console)"); // 設定顯示訊息  
        console.log("GET /posts?_limit=5 回傳資料:", data);  
  
        clearList(); // 清空文章列表  
        for (let i = 0; i < data.length; i++) {  
            addPostToList(data[i]);  
        }  
    } catch (error) {  
        setStatus("GET 失敗：" + error.message, false);  
        console.error("GET 錯誤:", error);  
    }  
};
```

Refer to: [js-14-fetch+await+async.html](#)

Supplementary:

Using Google Spreadsheet as a Backend

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□ Refer to: js-15-google-sheet.html

	A	B	C	D	E	F	G
1	姓名	統率	武力	智力	政治	總和	生年
2	諸葛亮	98	38	100	95	331	181
3	司馬懿	97	63	98	91	349	179
4	賈詡	88	48	97	85	318	147
5	龐統	86	34	97	85	302	178
6	周瑜	95	71	96	86	348	175
7	郭嘉	84	15	96	84	279	170
8	陸遜	94	69	95	87	345	183
9	荀彧	62	14	95	99	270	163
10	法正	84	47	94	78	303	176
11	荀攸	73	26	94	88	281	157
12	徐庶	87	64	93	80	324	178
13	田豐	72	29	93	87	281	162
14	李儒	61	26	93	75	255	150
15	曹操	99	72	92	91	354	155

武將名	統率	武力	智力	政治	總和
諸葛亮	98	38	100	95	331
司馬懿	97	63	98	91	349
賈詡	88	48	97	85	318
龐統	86	34	97	85	302
周瑜	95	71	96	86	348
陸遜	94	69	95	87	345
法正	84	47	94	78	303
徐庶	87	64	93	80	324
曹操	99	72	92	91	354
魯肅	85	56	92	90	323
鐘會	83	50	91	78	302
姜維	92	89	90	66	337
鄧艾	93	87	89	81	350



Supplementary:

Using Google Sheet to Upload/Display Images

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□ Refer to: [js-16 -image-upload-display.html](https://js-16-image-upload-display.html)

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* 表示必填問題

電子郵件 *

shangpin.ma@gmail.com

照片描述

Scrum

您的照片 *

可上傳 1 個支援的檔案 (image)，大小上限為 10 MB。

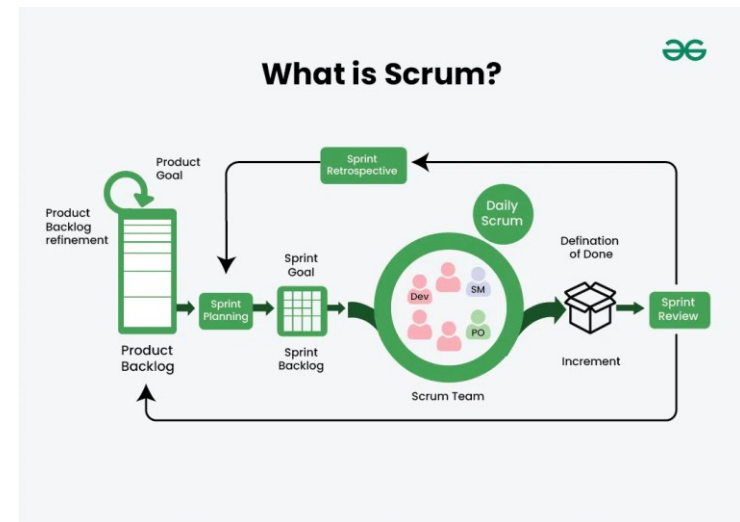
What-is-Scrum.w... X

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請勿利用 Google 表單送出密碼。

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albert@ntou.edu.tw	笑臉
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2024/11/26 上午 7:46:03	(圖片連結)



Any Question?

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